Exercises
Mastering Practical Closed-loop Control

Exercise 1: Participant acts as controller

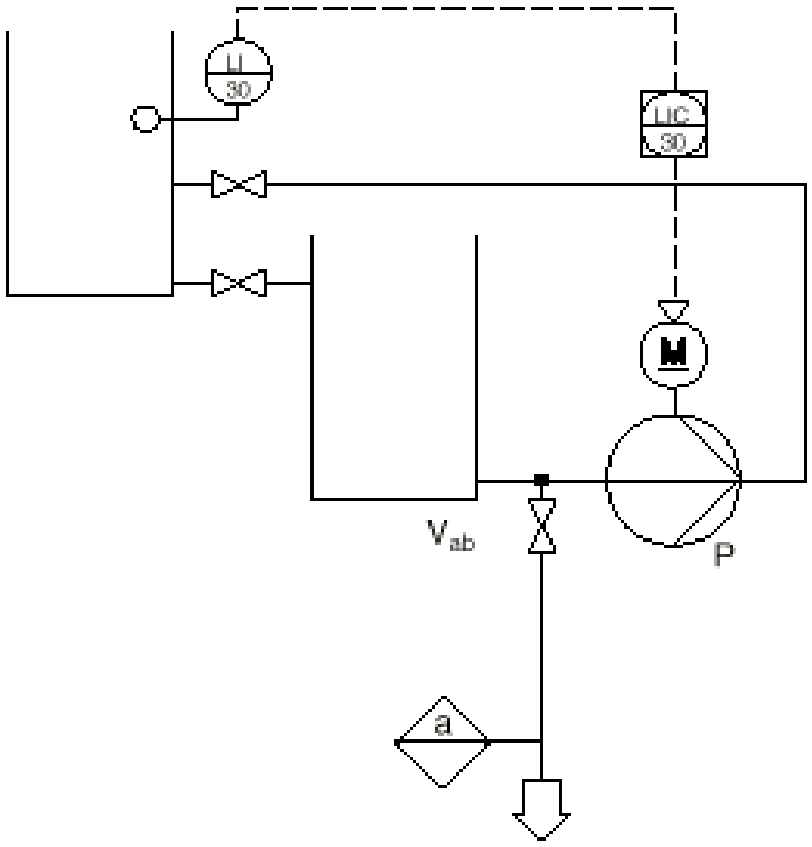
Objective
You understand the basic principles of closed-loop control and know the relevant terminology and abbreviations:

- controller
- controlled system
- set point
- process value (actual or measured value)
- controller output (manipulated variable)
- control parameters
- disturbance
- deviation

Material required
Level control station 170 669
or
Compact system 170 666
Foils - explanation of technical terms

Sequence

Set up the system as shown in the PI flow diagram:



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Fill the lower tank with water.
Define a set point on the upper tank - e.g. 100 mm.
Switch on the main power (On/Off)
Switch on the pump. The pump must run constantly during the exercise.

The aim is for one participant to go to the set point in the upper tank and hold it stable. This is done by operating the input valve.

As second participant creates disturbances by operating the valve between the upper and lower tank.

During the exercise the instructor will explain all of the terms listed above.

Conclusion
By experimentation you have learned the principles of closed-loop control and the terminology used to express them.

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Exercise 2: Demonstration of P, I and D components

Objective
You understand the Proportional, Integral and Differential components of a controller and how they work together.

Material required
Video of demonstration - part of the Turnkey Course Kit
Foils: Examples of P, PI and PID controllers
PC with InTouch and PCS application
Flow rate control station 170 668
OR
Compact system 170 666

Sequence
see video (in preparation!)

Conclusion
You now understand the difference between the explanations in theoretical books on closed-loop control and the way the P, I and D components work in practice.

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Exercise 3: Operating modes of an industrial controller

Objective
You can operate an industrial controller selecting the modes

- manual
- automatic
- configuration (setting parameters)

as required.

Material required
Flow rate control station 170 668
OR
Compact system 170 666
Technical documentation: Operating Manual for controller
Foils: Principles of Industrial Controllers

Sequence
Ensure that the controller is pre-configured for flow control (as in table below) .

Structure	**Standard**	
Input 1	Frequency	
	fre:	1000
Scaling	PVh	18.75
	PVl	0.00
Setpoint limits	SPh	18.75
	SPl	0.00
Alarm mode	AL	ABS
Alarm limits	PV+	18.75
	PV-	0.00
	HY	1.00
Filter	FG1	20.00
Controller	Kp1	10.00
	Tr	0.5
	Td	0.0
	CO0	0.0
Output	continuous	0-10V
Correcting variables limit	Coh	i00.0
	COl	0.0
Direction of control action	Inv	no
Safety	Cos	0.0
Adaption control	Tune	off
	Adapt	off
Additional	Serial	local

The manual mode is used when starting up and shutting down in order to get the system near to the set point. This prevents violent reactions caused by a large difference in setpoint and process value. In this exercise a flowrate of 3l/min is to be maintained.

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1. Starting up and shutting down

Sequence
Switch on power.

Start up
Switch the controller to the manual mode.
Note: LED on manual/automatic must be off
Press the display button until the manipulated variable (Co)is shown in the first line.
Use the arrow keys to adjust the controller output until you get close to the setpoint (I in the second line is approximately 3.00).
Before switching to automatic mode change the setpoint to 3.00 as follows:
Press the display button until the set point (SP) is displayed in the first line.
Using the arrow keys to change the setpoint to 3.00 (as previously described by the instructor).
Switch to automatic mode (LED on)

Shut down
Switch to manual mode
Press the display button until the manipulated variable (Co)is shown in the first line.
Use the arrow keys to adjust the controller output to 0.00.

2. Configuring the controller (adjusting the PID parameters)
Start up again (as above) to a setpoint of 3.00.
Switch to automatic mode.
Switch to configuration mode (press keys “Select“ and “Enter“ simultaneously for 5 s)
Using select, go to the controller menu and change the reset time TR to 1.00s.
Quit the configuration mode

Repeat this operation to change the reset time back to 0.5s.
Before quitting the configuration mode, cause disturbances to the system by operating the valve.
Observe how the controller behaves during this operation.
Quit the configuration mode.

Conclusion
You are now familiar with the operating modes of an industrial controller and can switch between them safely.

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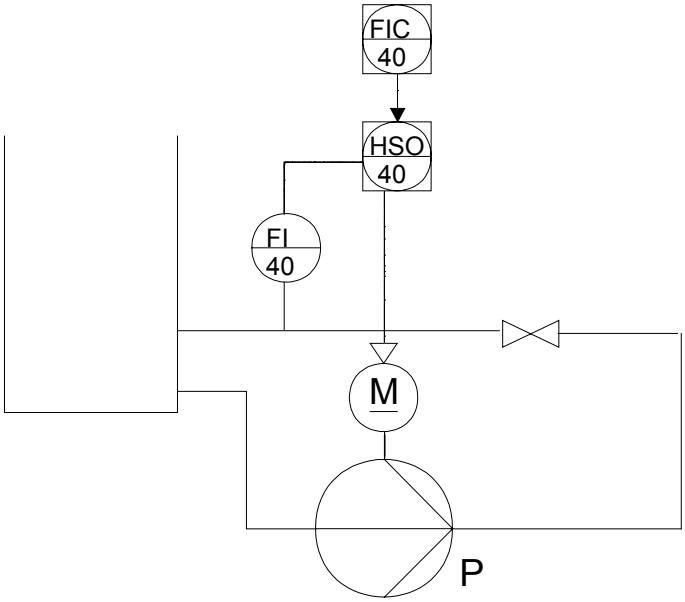
Exercise 4: Step-by-step automation

Objective
You can assemble and wire a basic closed-loop control system, integrate the controller and commission the system.

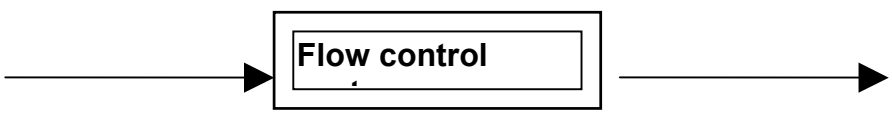
Material required

Profile plate	159 410
Tank	170 707
Power supply	local Didactic standard
Pump	170 712
Motor controller	170 698
Flow sensor	170 711
Controller	170 696
Voltage converter	167 001
Piping material	selection
Wiring material	sufficient
Tools	see list in Instructor's Guide
Foils	

Sequence
Introduction
Water is to be pumped in a circuit at a constant rate. Assemble the system according to the PI diagram (without electrical wiring)

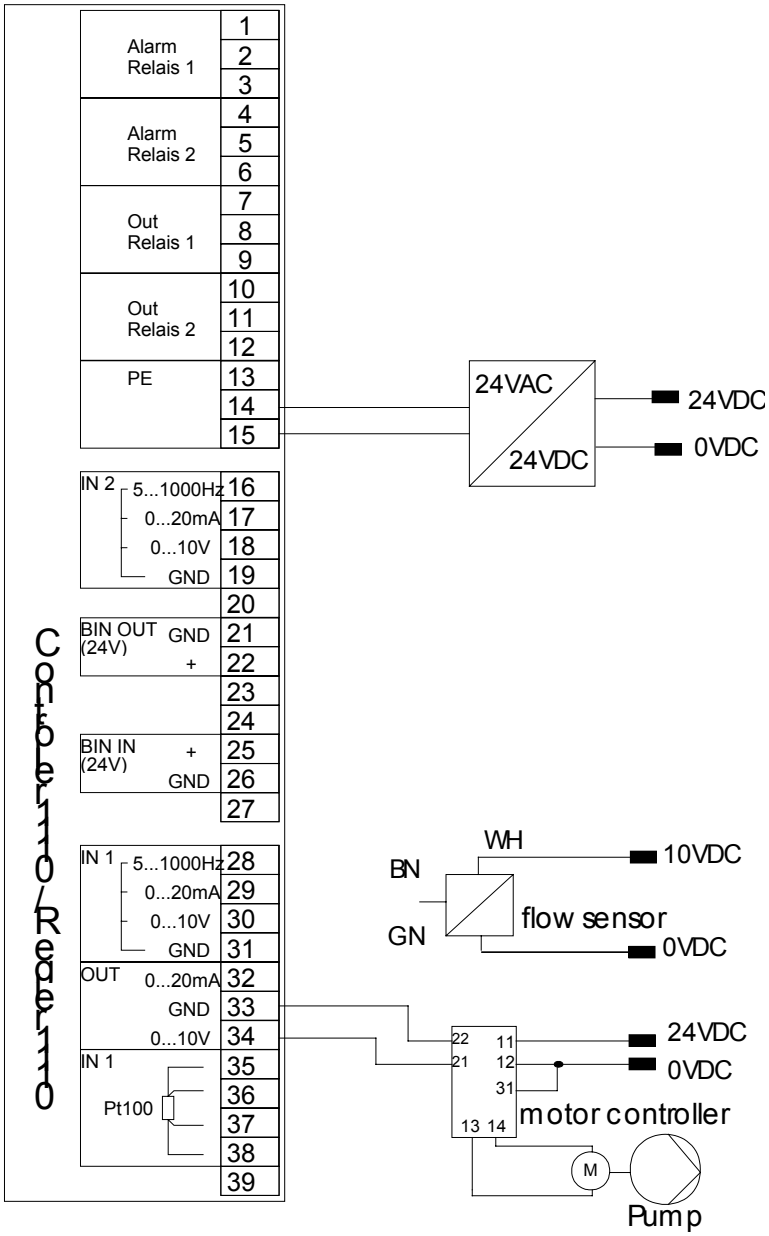


In your group discuss which input and output variables the flow control system has. Enter your Input and Output variable in the diagram below:



Input:_____ Output:_____

Complete the circuit diagram:



Wire up the system

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Configure the controller using the table below:

Structure	**Standard**	
Input 1	Frequenc y	
	fre:	1000
Scaling	PVh	18.75
	PVl	0.00
Setpoint limits	SPh	18.75
	SPl	0.00
Alarm mode	AL	ABS
Alarm limits	PV+	18.75
	PV-	0.00
	HY	1.00
Filter	FG1	20.00
Controller	Kp1	10.00
	Tr	0.5
	Td	0.0
	CO0	0.0
Output	continuou s	0-10V
Correcting variables limit	COh	i00.0
	COl	0.0
Direction of control action	Inv	no
Safety	COs	0.0
Adaption control	Tune	off
	Adapt	off
Additional	Serial	local

Commission the system
Once the system has been commissioned correctly conduct the following system analysis:

What is the max. flowrate? Flow max: _____
What is the min. flowrate? Flow min: _____

What is the range of the system response time? (tick your selection):

☐ 0-50ms
☐.50ms-5s
☐.5s-1min
☐.1-5min
☐.5-60min

Conclusion

Although process control systems appeared difficult at the beginning, a step by step approach makes it easy to understand and ensures that the desired operation is achieved.

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Exercise 5: Ziegler Nichols Method

Technical controllers are component parts of automation systems, whose main task is that of process stabilisation. Different methods are used in practice to set the control parameters.

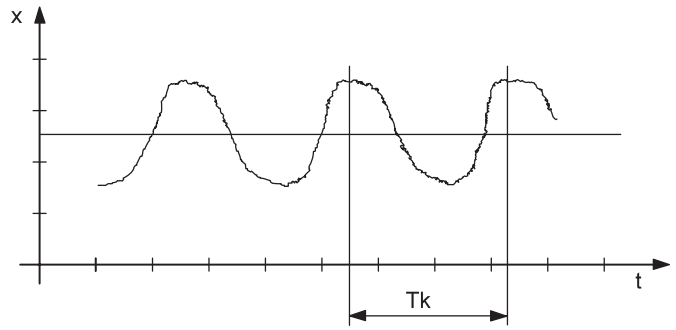
Exercise 1

- Setting of control parameters using the Ziegler/Nichols method
  - Determine the operating point:
    Set the controller to `Manual' and determine the possible control range by means of changing the correcting variable. Select the operating point so that the controller has sufficient reserve in both modulation directions, (e. g. in the centre of the control range).

Correcting variable	**Actual value**
Y = 0%	X1 =
Y = 100%	X2 =

Selected operating point (X1 + X2) / 2:
W =

  - Configure the controller as a P controller: To do this, set T_n at the highest possible value (9999.) and T_v at 0.
  - Determine the critical amplification K_r (stability limit, closed control loop is in the process of carrying out continuous oscillation), by analysing small setpoint step-changes around the operating point after each newly set amplification. This determines the critical amplification factor K_{KR} and the period of oscillation T_k of this continuous oscillation.



K_p	closed control loop oscillates	closed control loop does not oscillate
10	Lower K_p	Increase K_p

Note:
The flow and filling level control systems have been selected in such a way that the oscillating process can be observed without any auxiliary equipment (just eyes and ears).

Controlled system parameter	
K_{KR}	
T_K	

- The control parameters K_p, T_n, and T_v are determined according to controller type with the help of the table.

Controller type	**K_P**	**T_n**	**T_v**
P	0,5 K_{KR}	–	–
PI	0,4 K_{KR}	0,85 T_K	–
PID	0,6 K_{KR}	0,5 T_K	0,12 T_K

- Selected parameters:

Controller type	**K_P**	**T_n**	**T_v**
P			
PI			
PID			

- Verify the determined control parameter of a PI controller by carrying out a setpoint step-change from approx. 20% to 60% of the modulation range.
- What is the closed control loop behaviour if you increase (e. g. double) T_n?
- What is the closed control loop behaviour if you increase the amplification (e. g. by 30%)?

In addition, also carry out the setpoint step-changes.

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Exercise 6: Autotune and adaptation

Objective
You can perform an autotune and an adaptation and evaluate the results.

Material required

Flow rate control station	170 668
OR	
Compact system	170 666
Technical documentation:	Operating Manual for controller
Foils	Autotune and Adaptation

Sequence
The exercise will show you the two automatic methods of configuring the controller's parameters and their limitations.

Autotune
Configure the controller using the table below:

Structure	Standard	
Input 1	Frequency	
	fre:	1000
Scaling	PVh	18.75
	PVl	0.00
Setpoint limits	SPh	18.75
	SPl	0.00
Alarm mode	AL	ABS
Alarm limits	PV+	18.75
	PV-	0.00
	HY	1.00
Filter	FG1	20.00
Controller	Kp1	10.00
	Tr	0.5
	Td	0.0
	CO0	0.0
Output	continuous	0-10V
Correcting variables limit	COh	i00.0
	COl	0.0
Direction of control action	Inv	no
Safety	COs	0.0
Adaptation control	Tune	on
	Adapt	off
Additional	Serial	local

Switch to manual mode
Set the controller output to 0.00 using the arrow keys
Change the setpoint to 0.00 and enter
Switch to automatic mode
Change the set point to 3.00
Press Enter

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After the tuning sequence
read out the controller's parameters:
Switch to configuration mode (press keys "Select" and "Enter" simultaneously for 5 s)
Using select, go to the controller menu and read out the parameters Kp and Tr and enter them in the table below.
Quit the configuration mode

Table:

	Value	Remarks
Kp		
Tr		

Adaptation
Configure the controller using the table below:

Adaption control		
	Tune	off
	Adapt	on
	SP	yes
	PV	no

Switch to manual mode
Set the controller output to 0.00 using the arrow keys
Change the setpoint to 0.00 and press enter
Switch to automatic mode
Change the set point to 3.00
Press Enter

After the adaptation sequence
Adjust the setpoint between 2.00 and 3.00 several times until the controller stabilises.

Read out the controller's parameters:
Switch to configuration mode (press keys "Select" and "Enter" simultaneously for 5 s)
Using select, go to the controller menu and read out the parameters Kp and Tr and enter Them in the table below.
Quit the configuration mode

Table:

	Value	Remarks
Kp		
Tr		

Within your group find an explanation for the way the system responded in each case.

Conclusion
While Autotune and Adaptation are a help to find suitable parameters, knowledge of closed-loop control is essential to evaluate the results. It is particularly useful when different parameters at different setpoints (non-linear systems).

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Exercise 7: Alarm procedures

Objective
You know how to determine and set alarm limits to protect the system

Material required
Level control station 170 669
OR
Compact system 170 666
Technical documentation: Operating Manual for controller

Sequence
Configure the controller using the table below:

Structure	**Standard**	
Input 1	0-20mA	
	rad	no
Scaling	PVh	50.00
	PVl	-20.00
Setpoint limits	SPh	50.00
	SPl	0.00
Alarm mode	AL	ABS
Alarm limits	PV+	50.00
	PV-	0.00
	Hy	1.00
Filter	FG1	20.00
Controller	Kp1	28.00
	Tr	5.0
	Td	0.0
	CO0	0.0
Output	continuous	0-10V
Correcting variables limit	COh	i00.0
	COl	0.0
Direction of control action	Inv	no
Safety	COs	0.0
Adaptation control	Tune	off
	Adapt	off
Additional	Serial	local

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1. Absolute alarm mode
The absolute alarm sets upper and lower limits for the process value in the system.

Task: Set the alarm to trigger at 100mm (lower limit) and 200mm (upper limit)

In your group, define the upper and lower limits for level. Read out the process value when the level is at 100mm and at 200mm.

Enter these values in the table below

Alarm limit	**Level in mm**	**Process value**	
Lower limit	100	PV = _____	use this value for X1-
Upper limit	200	PV = _____	use this value for X1+

Enter the values in the controller as PV+ and PV-.
Adjust the system so that the alarm is triggered.

2. Relative alarm mode
The relative alarm sets the permitted deviation above and below the setpoint in the system.

Task: Set the alarm to trigger at 10 mm above the setpoint and 20 mm below the setpoint.

Change the alarm mode to relative:

Alarm mode	AL	REL

Read out the process value for 10mm above and 20mm below the setpoint.
Enter these values in the table below

Alarm limit	**Level in mm**	**Process value**	
Lower limit	SP +10mm	PV =	use this value for -
Upper limit	SP -20mm	PV =	use this value for

Enter the values in the controller as PV+ and PV-.
Adjust the system so that the alarm is triggered.
Check the limits using 2 different setpoints

Conclusion
Industrial controllers offer two alarm functions.

Absolute alarms are used to ensure that systems remain within the permitted absolute range in order to protect life, the system and the medium.

Relative alarms are used to inform the operator when the process value moves too far from the current setpoint.

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Exercise 8:Optimising the controlled system

Objective
You can improve the physical characteristics of a controlled system by changing the system in order to

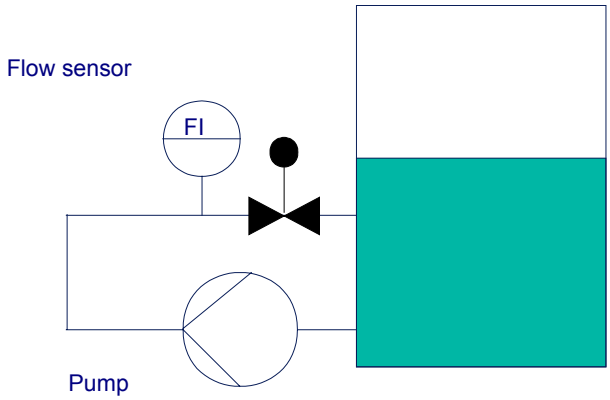
- eliminate dead time
- minimise delay times
- increase reset times

Material required

Flow rate control station	170 668
OR	
Compact system	170 666
Pressure control system:	(no number allocated)
Technical documentation:	Operating Manual for controller
Accessories:	Additional piping material
Foils:	Optimising closed-loop control behaviour

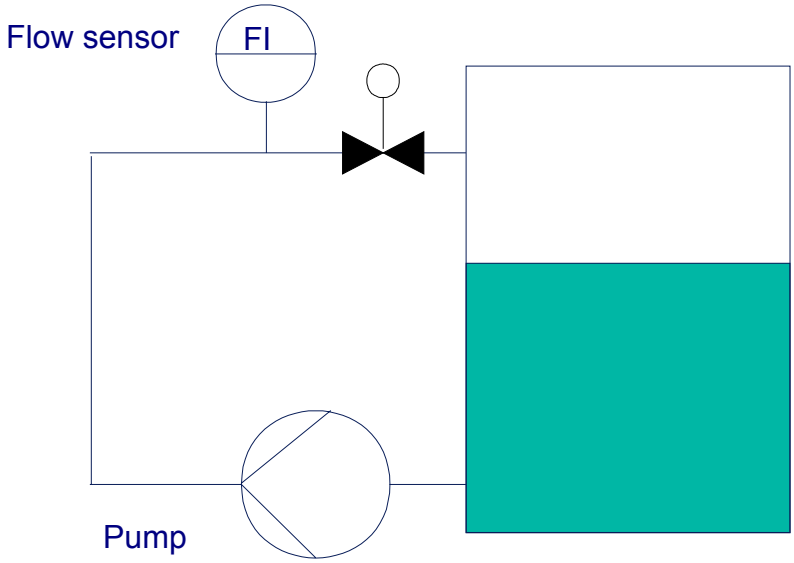
Sequence

Flowrate 1
Set up the system as follows and carry out the sequence below:



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Flowrate 2

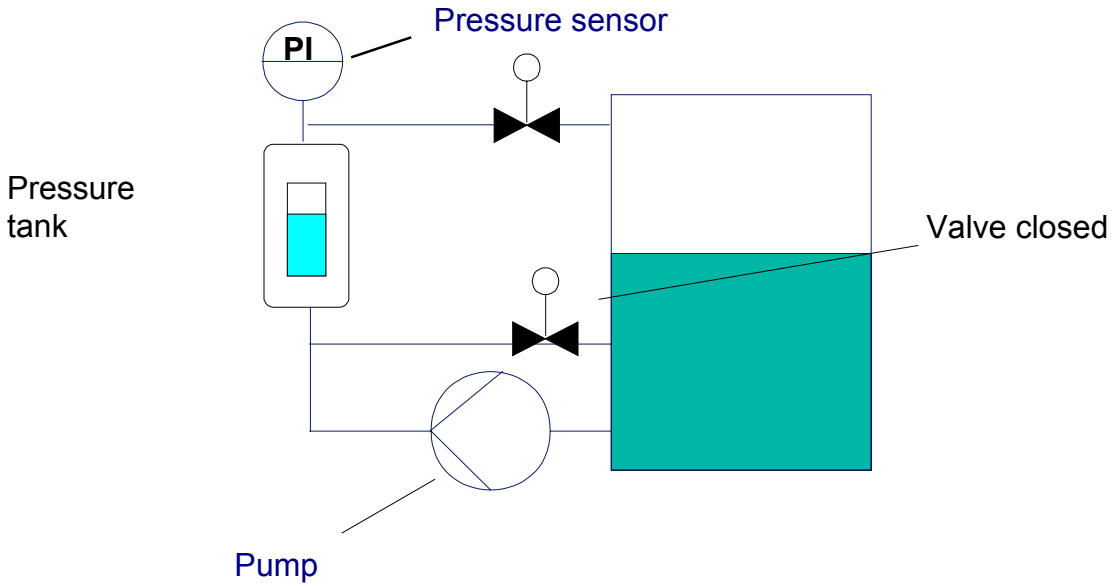


Sequence
Using the Ziegler-Nichols method, determine suitable parameters for the system
Set the parameters
Fill in the table below.
Change the setpoint in automatic mode from 0.00 to 3.5 and note how the system reacts.
Discuss the reasons for the systems response within your group.

Table:

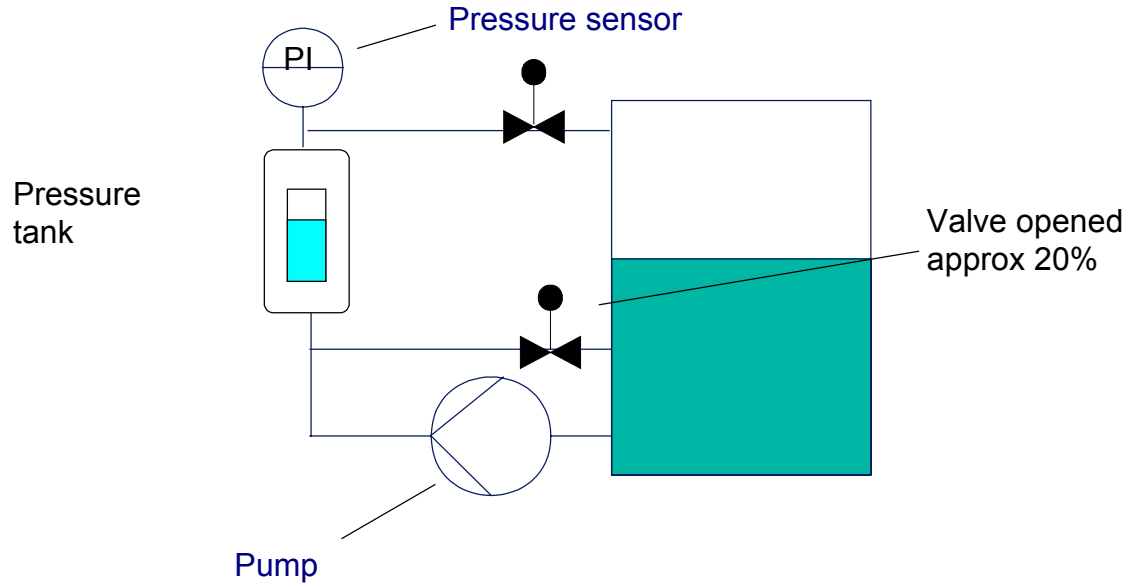
	Flowrate A	Flowrate B
Parameters		
P		
I		
D		
Remarks		

Pressure 1
Set up the system as follows and carry out the sequence below:



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Pressure 2

Set up the system as follows and carry out the sequence below:



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Sequence

Using the Ziegler-Nichols method, determine suitable parameters for the system
Set the parameters
Fill in the table below.
Change the setpoint in automatic mode from 0.00 to 50mbar and note how the system reacts.

Discuss the reasons for the systems response within your group.

Table:

	Pressure A	**Pressure B**
Parameters		
P		
I		
D		
Remarks		

Conclusion

In any control there is a controller and the controlled system. When you are optimising the system, make sure you pay attention to both of them. If you have trouble finding good PID parameters, take a hard look a the system and see if you can improve it.